2D One-/Two-Target Gating Mechanism Report

one-target reset to a shared symmetric baseline with balanced asymmetric comparisons

Headline Result

This report still preserves the March 16 source chain `membrane_gating_game` → `two_target_gating_framework` → `one_two_target_deepening_work`, but the one-target branch has now been reset onto a more stable configuration logic:

- the **standard configuration** is the shared symmetric baseline `sym_shared`, not an extreme asymmetric point;
- the manuscript separates two distinct notions of asymmetry: **top/bottom membrane asymmetry** and **same-membrane directional asymmetry**;
- **rollback** is retained as the main one-target discrete mechanism, but now in a *gate-free* form: after leaving the start basin A, does the trajectory later re-enter A;
- the real-set **x-gate** remains, but only as a time-anchor line for the $N/P/Q$ mechanism;
- **loss of bimodality** is defined strictly as `phase=0`; `phase=1` means weakening, not disappearance of the second peak.

1 Source Chain and Current Role

Why the one-target branch needed to be rewritten again. The previous canonical integration had already absorbed the 6 downloaded files into shared code and a standard report layout, but the one-target main language was still not stable enough: extreme $\kappa = 0$ points were too exhibition-like, and rollback remained too tightly tied to the $x = X_g$ gate. The present revision resets the one-target section to a more geometric and more repository-native language.

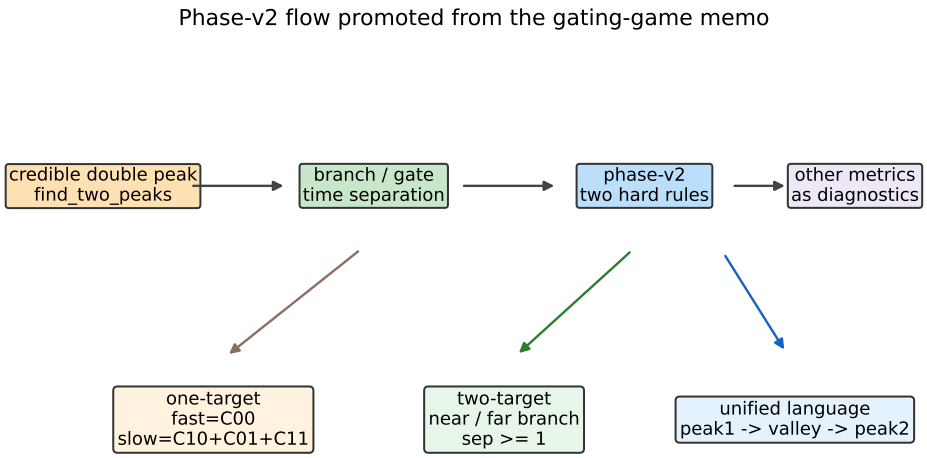


Figure 1: The phase-v2 / gating-game source sketch. The current canonical report keeps the source chain, but no longer lets the older gate language dominate the one-target main text.

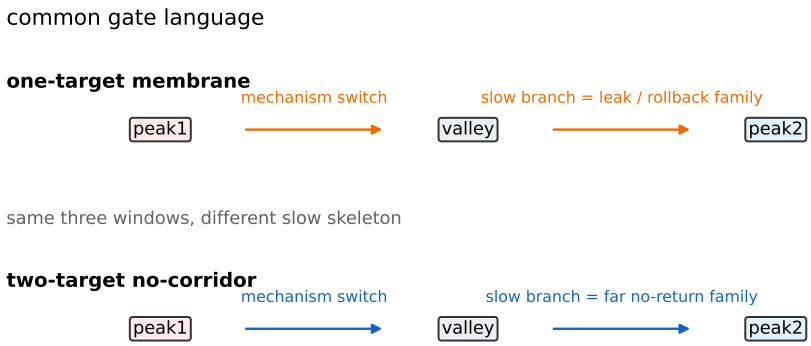


Figure 2: Unified mechanism ladder. The two-target branch still uses the gate-lifted family language, while the one-target branch is now organized around a shared symmetric baseline, gate-free rollback classes, and x-gate timing diagnostics.

2 The one-target standard configuration and the two symmetries

What the standard configuration is. The one-target standard configuration is now fixed as

$$\texttt{sym_shared}: \qquad \kappa_{\text{top}} = \kappa_{\text{bottom}} = k_{\text{ref}} = 0.002,$$

which simultaneously satisfies the directional symmetry condition

$$\kappa_{c\to o} = \kappa_{o\to c} = 0.002.$$

So `sym_shared` is the unique shared baseline for both symmetry notions.

Why the canonical representatives are no longer extreme. Extreme points remain useful as controls or phase-loss boundaries, but they are no longer used as the main representatives. The current representatives are selected by a fixed scan rule:

- prefer `phase=2`;
- prefer interior points with both parameters strictly positive;
- prefer points close to fixed balanced anchors, while still keeping strong peak separation;
- if the nominal anchor is not itself phase-2, select the nearest phase-2 interior point in the same directional class.

As a result, the non-symmetric representatives are now balanced comparison cases rather than "one side completely shut" showcase points.

Table 1: Canonical one-target representative cases. The table explicitly distinguishes the standard configuration from comparison configurations and records the actual parameters, phase, and separation values.

case	role	parameters	phase	sep	peak2 rolll
shared symmetric baseline	standard	$(\kappa_{c\to o}, \kappa_{o\to c}) = (0.0020, 0.0020)$	2	1.528	L0R1
top/bottom balanced asymmetry	comparison	$(\kappa_{top}, \kappa_{bottom}) = (0.0020, 0.0005)$	2	1.689	L0R1
directional easy-out / hard-return	comparison	$(\kappa_{c\to o}, \kappa_{o\to c}) = (0.0020, 0.0005)$	2	1.614	L0R1
directional hard-out / easy-return	comparison	$(\kappa_{c\to o}, \kappa_{o\to c}) = (0.0005, 0.0020)$	2	1.737	L0R1

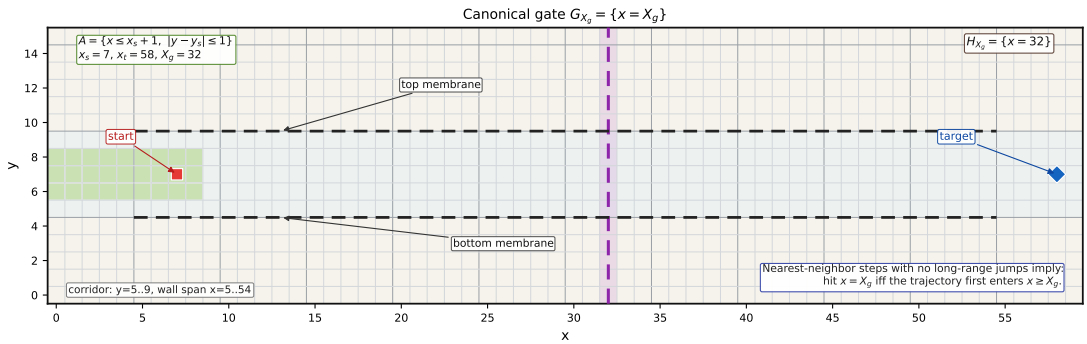


Figure 3: Standard one-target geometry. The panel is drawn for the shared symmetric baseline and marks the lattice, corridor, start basin A, target, the true membrane interfaces, and the canonical x-gate $G_{X_g} = \{x = X_g\}$.

What the two asymmetry notions mean. The one-target branch now keeps the two physically different symmetry lines separate:

1. **top/bottom symmetry**: whether the upper and lower membranes are the same;
2. **same-membrane directional symmetry**: whether `corridor`→`outside` and `outside`→`corridor` have the same permeability on a given membrane.

The first is scanned in the `tb` plane $(\kappa_{\text{top}}, \kappa_{\text{bottom}})$, and the second in the `dir` plane $(\kappa_{c\to o}, \kappa_{o\to c})$. They are numerically and physically different questions.

3 Gate-free rollback classes and x-gate timing classes

Why rollback is now gate-free. Rollback is now defined in a genuinely gate-free way: after a trajectory leaves the start basin A, does it later re-enter A? Under that definition the four rollback classes

$$L0R0, \quad L0R1, \quad L1R0, \quad L1R1$$

are mutually exclusive and exhaustive:

- $L0R0$: no leak and no return to A;
- $L0R1$: no leak but a return to A after leaving it;
- $L1R0$: membrane leak but no later return to A;
- $L1R1$: membrane leak and later return to A.

What the x-gate still does. The real-set x-gate is retained, but only for one purpose: to timestamp the *first* leak relative to

$$G_{X_g} = \{x = X_g\}.$$

This yields the timing classes

$$N = \text{NoLeak}, \qquad P = \text{first leak before gate}, \qquad Q = \text{first leak after gate}.$$

Under the present nearest-neighbor lattice dynamics, hitting $x = X_g$ is equivalent to first entering $x \ge X_g$, so only the single canonical x-gate line remains in the main text.

4 Top/bottom parameter scan

What this scan answers. This 15×15 scan over

$$\kappa_{\text{top}}, \kappa_{\text{bottom}} \in \{0, 0.00025, \ldots, 0.005\}$$

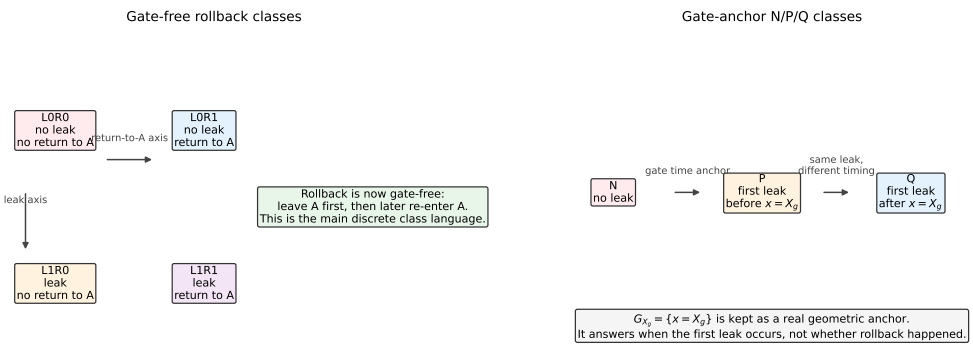


Figure 4: One-target class explainer. The rollback 4-class system answers "did leak occur" and "did the path return to A"; the $N/P/Q$ system answers "when did the first leak occur relative to the x-gate." The two are parallel diagnostics rather than competing definitions.

asks how upper/lower membrane imbalance changes double-peak strength, phase-0 loss, and the balance between rollback and x-gate timing effects.

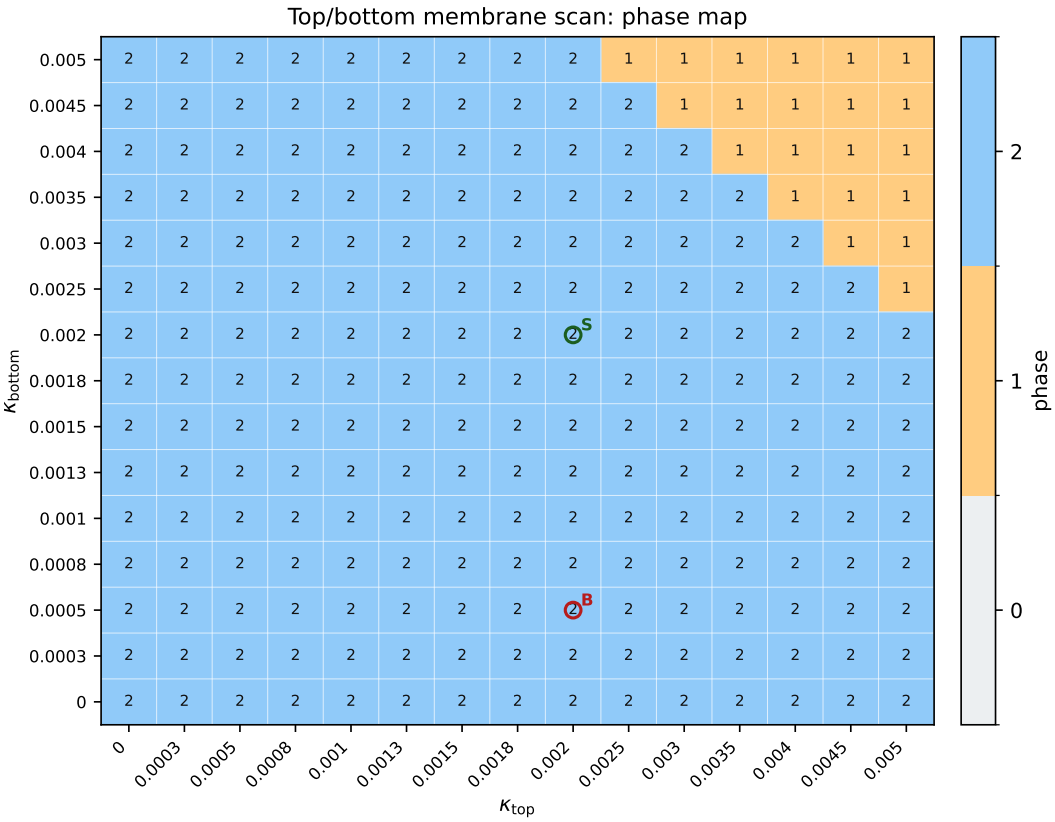


Figure 5: Top/bottom parameter-plane phase map. The shared symmetric baseline and the balanced top-open comparison point are explicitly marked.

5 Same-membrane directional scan

Why this scan is kept separate. The top/bottom scan alone cannot answer whether the same membrane is easy to exit but hard to re-enter, or the reverse. That directional issue is therefore scanned explicitly in

$$\kappa_{c\to o}, \kappa_{o\to c} \in \{0, 0.00025, \ldots, 0.005\}.$$

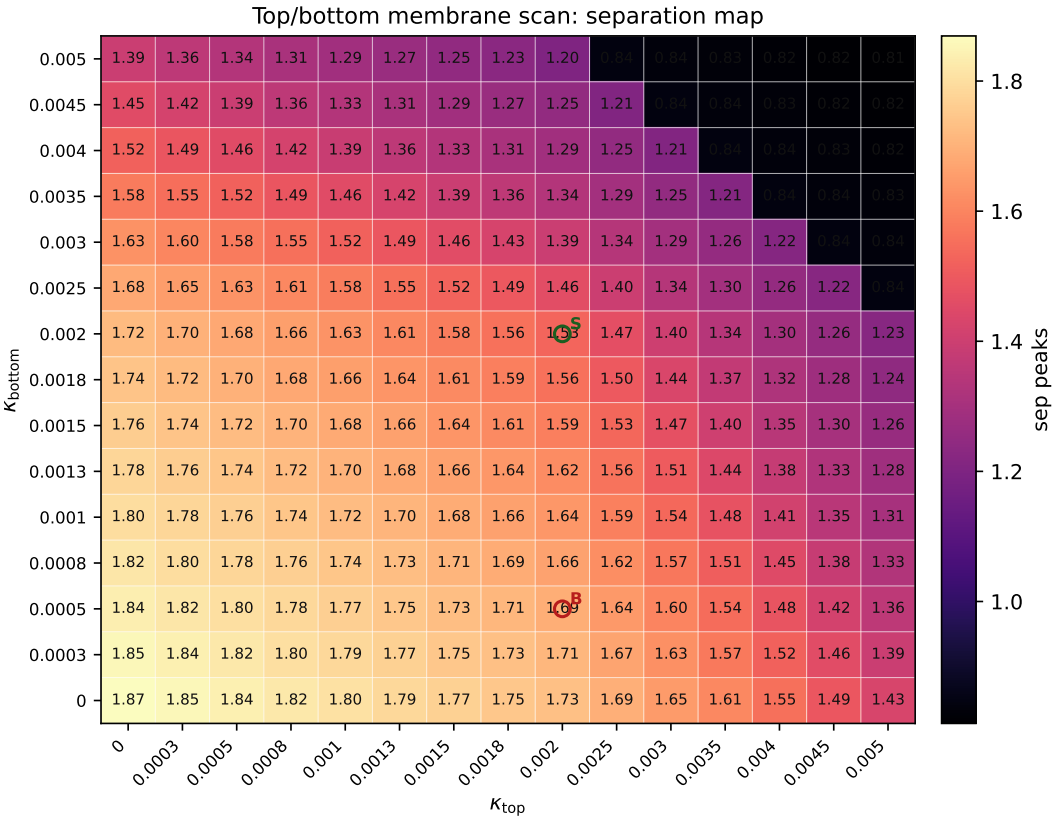


Figure 6: Peak-separation map on the same top/bottom plane. The color variation shows that moving away from the shared symmetric baseline first weakens and eventually can destroy the double-peak structure.

Table 2: x-gate scan summary. For each canonical case the table reports dominant-N counts and the ranges of P/Q/rollback across the full integer X_g scan.

case	scan range	dom. N	P range	Q range	rollback range
directional hard-out / easy-return	9..57	49	0.1%–2.8%	2.5%–5.2%	28.2%–76.4%
directional easy-out / hard-return	9..57	49	0.1%–8.0%	1.3%–9.1%	27.8%–77.2%
shared symmetric baseline	9..57	49	0.2%–8.6%	2.9%–11.4%	27.5%–77.0%
top/bottom balanced asymmetry	9..57	49	0.1%–5.4%	1.8%–7.2%	28.0%–76.8%

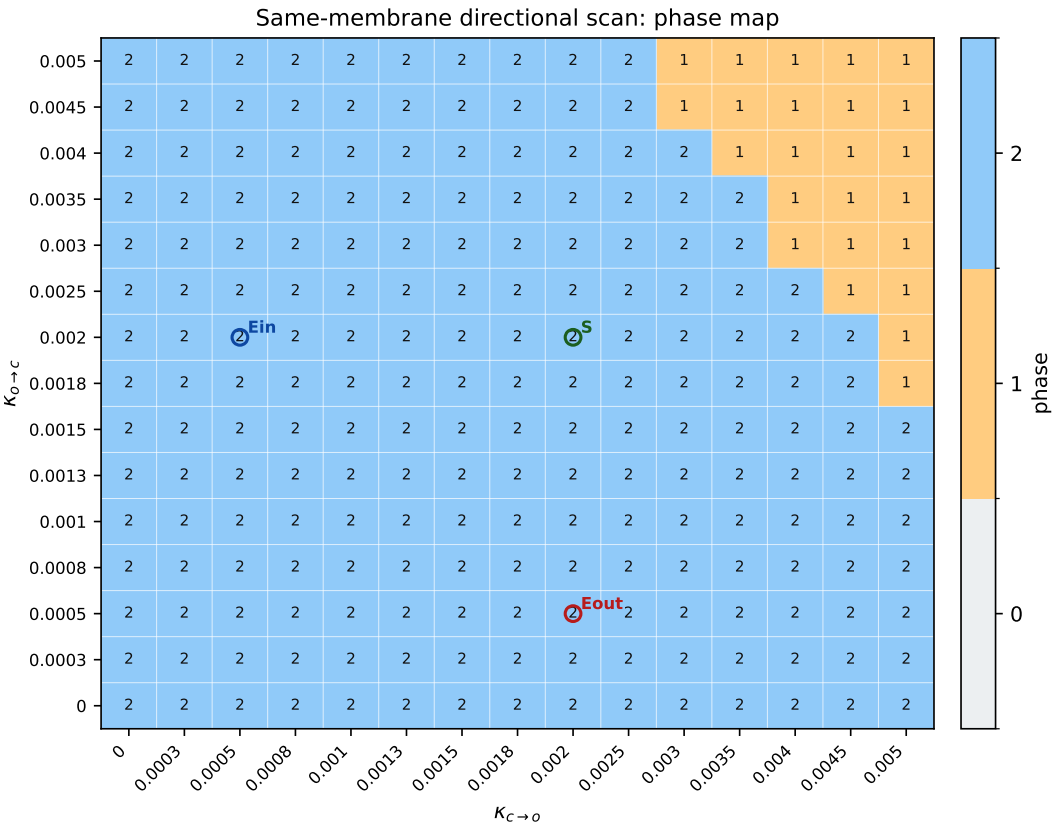


Figure 7: Phase map on the same-membrane directional plane. The shared symmetric baseline and the two directional comparison cases are marked explicitly.

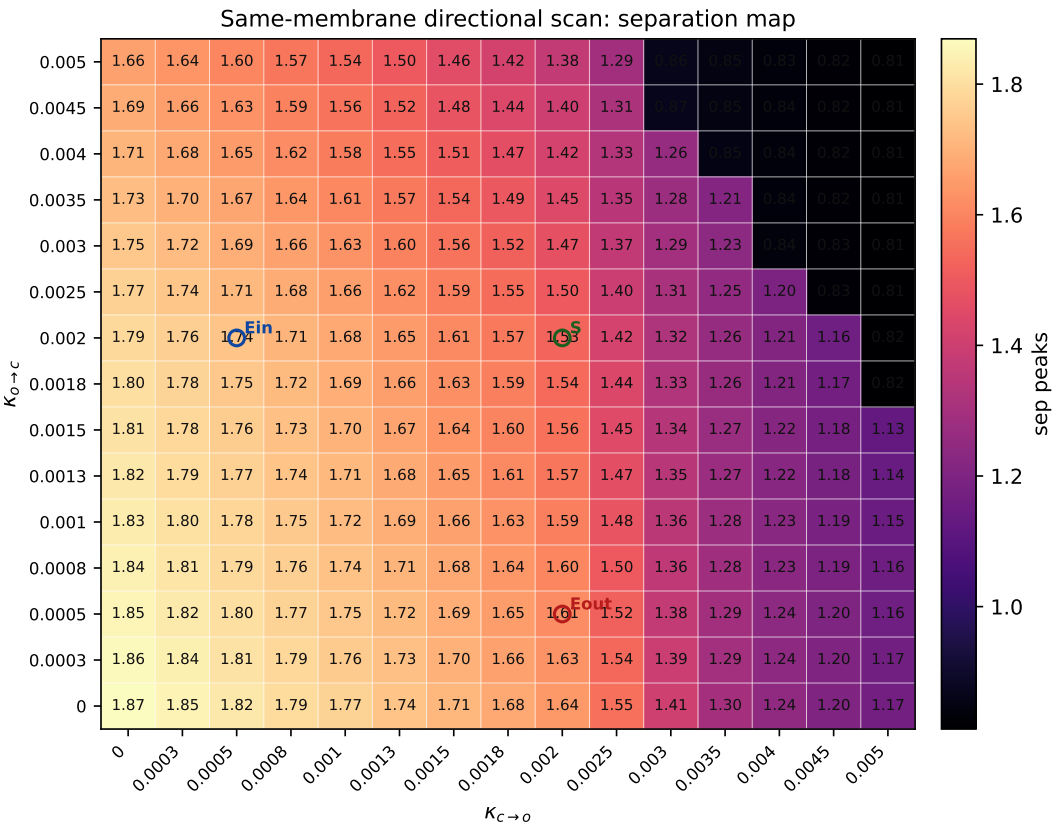
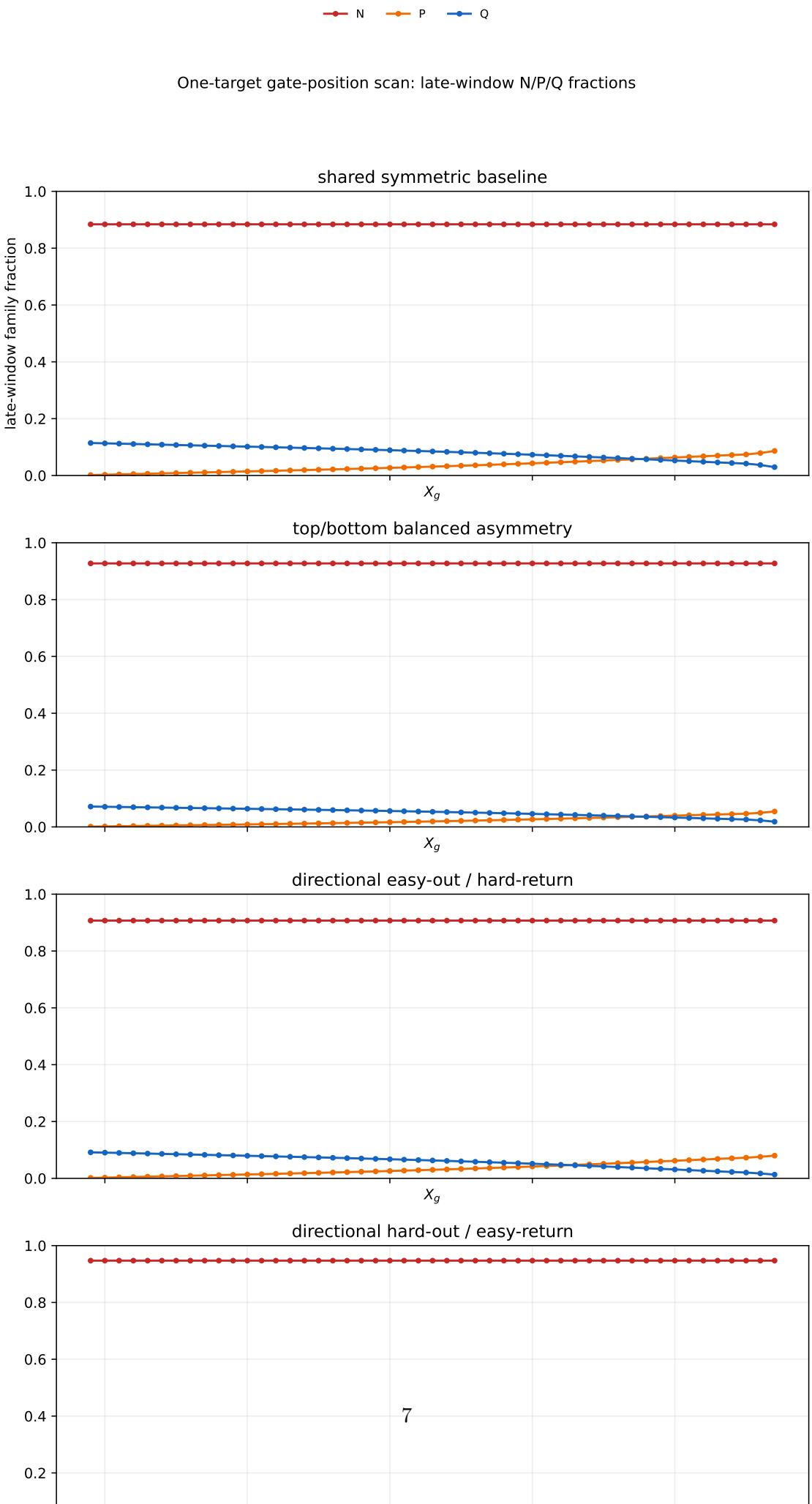
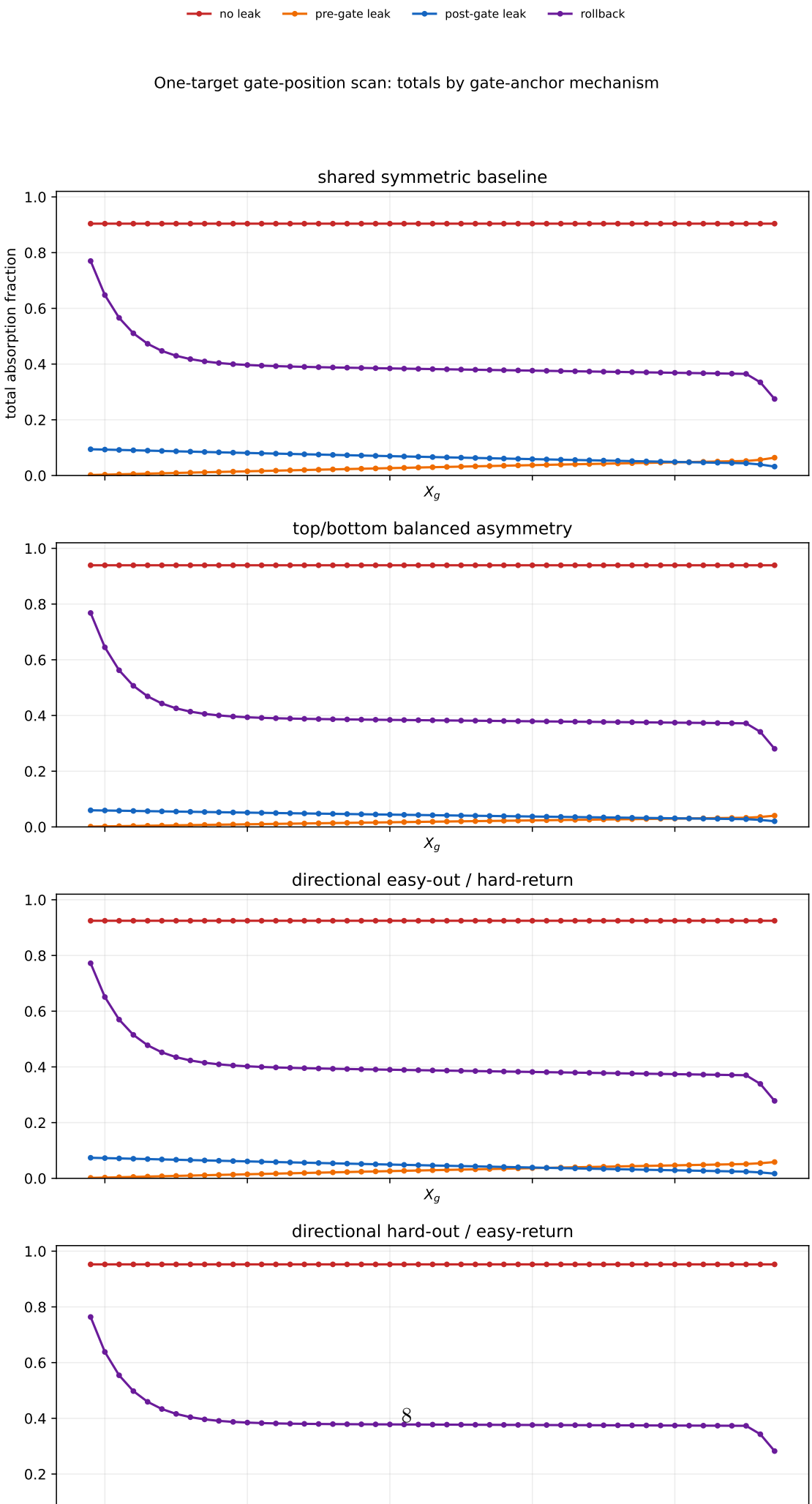


Figure 8: Peak-separation map on the directional plane. It shows directly how easy-out versus easy-in membranes alter the strength of the second peak.


N
P
Q
One-target gate-position scan: late-window N/P/Q fractions
shared symmetric baseline
late-window family fraction
1.0
0.8
0.6
0.4
0.2
0.0
x_g
top/bottom balanced asymmetry
1.0
0.8
0.6
0.4
0.2
0.0
x_g
directional easy-out / hard-return
1.0
0.8
0.6
0.4
0.2
0.0
x_g
directional hard-out / easy-return
1.0
0.8
0.6
0.4
0.2


no leak
pre-gate leak
post-gate leak
rollback
One-target gate-position scan: totals by gate-anchor mechanism
shared symmetric baseline
total absorption fraction
1.0
0.8
0.6
0.4
0.2
0.0
x_g
top/bottom balanced asymmetry
1.0
0.8
0.6
0.4
0.2
0.0
x_g
directional easy-out / hard-return
1.0
0.8
0.6
0.4
0.2
0.0
x_g
directional hard-out / easy-return
1.0
0.8
0.6
0.4
0.2

6 Start-position scan and phase-0 loss

Only `phase=0` counts as true loss of bimodality. In this report `phase=2` means a clear double peak, `phase=1` means weakening but not disappearance, and only `phase=0` means that the second peak is genuinely lost. So the question "when is the double peak lost" is interpreted strictly as a phase-0 question.

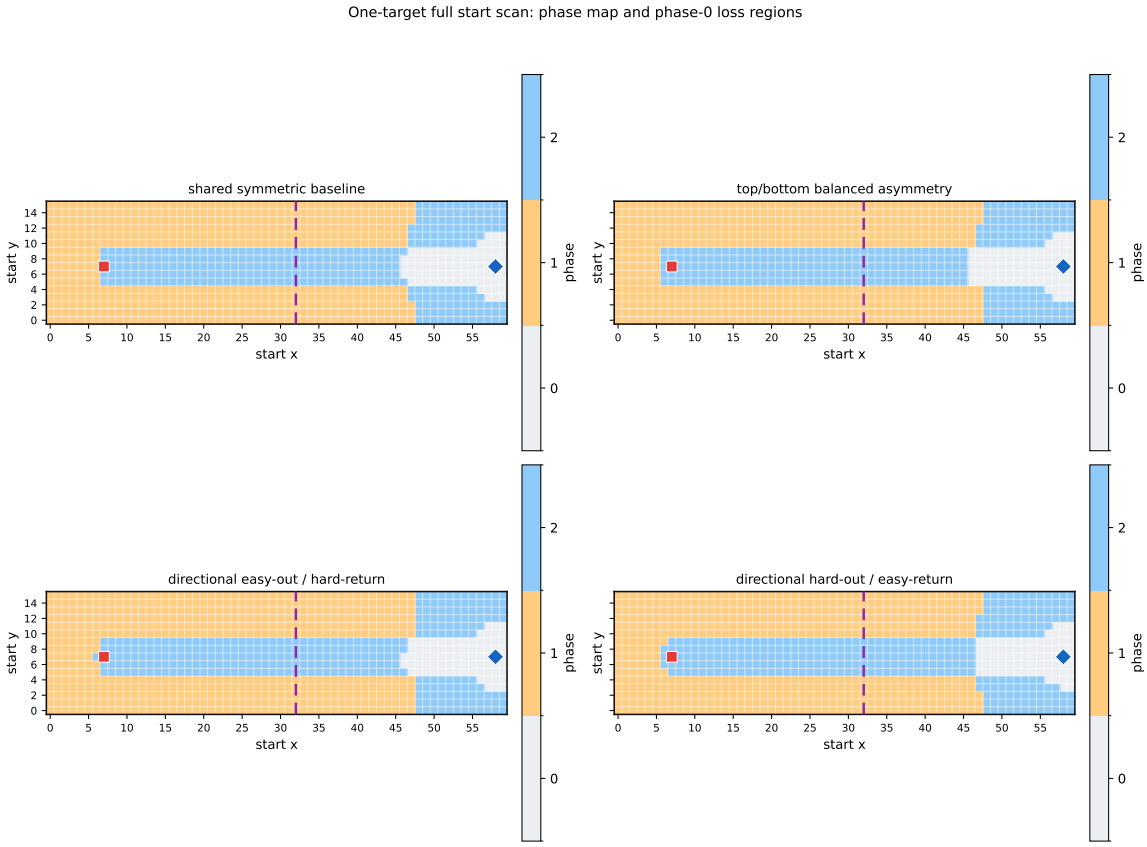


Figure 11: Four-case start-position phase atlas. This figure answers where phase-0 loss begins and how it expands for different membrane configurations.

Why there is no full leak-balance atlas here. The start-position section is meant to answer one question cleanly: where phase-0 loss appears first. The rebuilt canonical pipeline therefore keeps the full start atlas phase-only exact, while leak timing and rollback budgets are discussed in the parameter scans and representative window decompositions rather than being recomputed on every start grid point.

Table 3: Phase-0 loss summary. For each canonical case the report records a centerline onset, an off-axis loss point, and a target-adjacent collapse point.

case	phase-0 onset	off-axis loss	target-adjacent collapse	phase counts
shared symmetric baseline	(46, 7)	(47, 5)	(57, 7)	0:81/1:558/2:320
top/bottom balanced asymmetry	(46, 7)	(46, 5)	(57, 7)	0:83/1:555/2:321
directional easy-out / hard-return	(46, 7)	(47, 5)	(57, 7)	0:80/1:562/2:317
directional hard-out / easy-return	(47, 7)	(47, 5)	(57, 7)	0:77/1:555/2:327

7 Representative refinements: rollback, x-gate, and directional flux

Which mechanism is stronger. Under the present canonical rebuild, the stronger one-target discrete mechanism is the gate-free rollback system rather than the x-gate itself. The reason is simple: the rollback classes directly answer "did leak happen" and "did the path return to A," while the x-gate mainly functions as a timing anchor for the first leak.

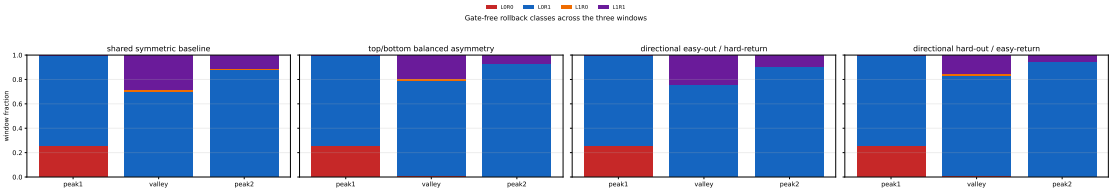


Figure 12: Representative three-window rollback decomposition. The quantity $L0R1 + L1R1$ is the direct measure of return-to-A weight in the valley and late windows.

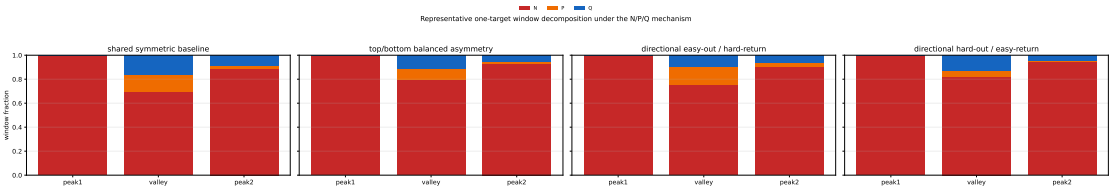


Figure 13: Representative three-window x-gate decomposition. Here $N/P/Q$ is used as a timing refinement, not as the primary discrete mechanism language.

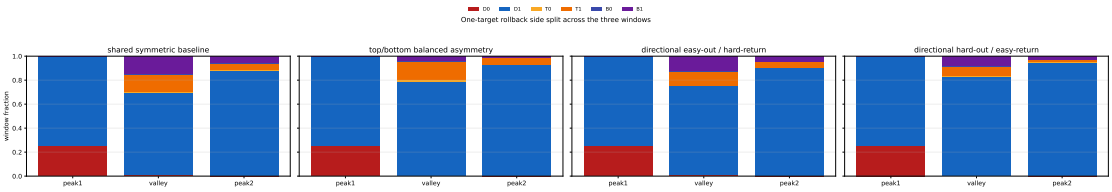


Figure 14: Side-aware rollback refinement. For the shared symmetric baseline, top/bottom is mainly a symmetry check; for the top/bottom asymmetric comparison case, it becomes a real side preference.

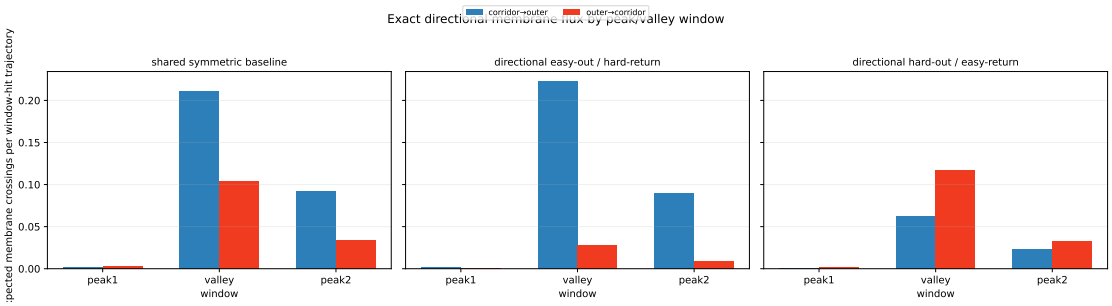


Figure 15: Directional window-flux diagnostics. This figure explains why easy-out versus easy-in membranes alter the second peak, rather than leaving the effect hidden at the parameter-name level.

8 Two-target: the gate-lifted family line stays unchanged

The two-target branch is numerically unchanged by this one-target rewrite. Its canonical conclusion remains that the late coarse family is stably `F_no_return`, not `F_rollback`. This section therefore keeps the canonical figures, scan slices, and verification tables without changing the underlying mechanism language.

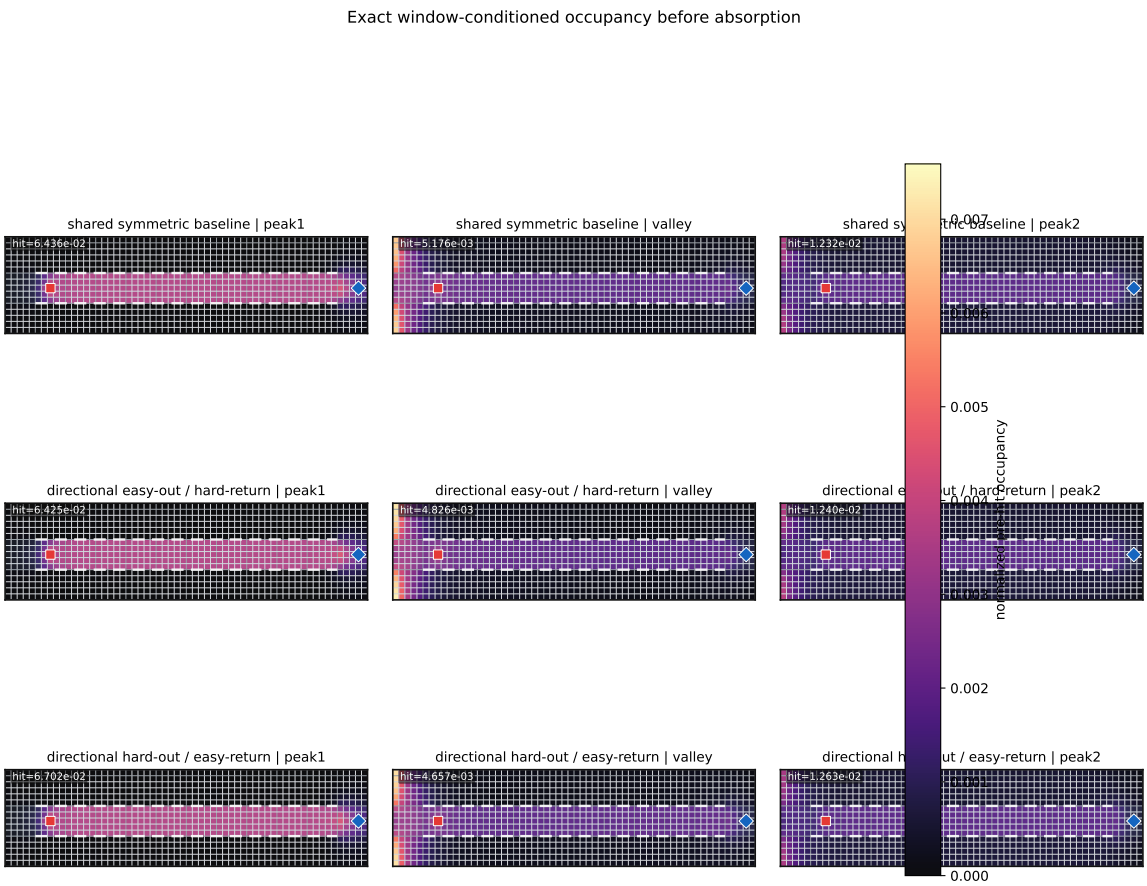


Figure 16: Representative window-occupancy atlas for the directional line. It puts the slow branch back onto the actual lattice geometry.

Table 4: Representative one-target recheck. The table reports phase, separation, peak2 rollback share, and the ranges of P/Q/rollback across the full X_g scan.

case	phase	sep	peak2 rollback share	P range	Q range	rollback
shared symmetric baseline	2	1.528	99.8%	0.2%–8.6%	2.9%–11.4%	27.5%–
top/bottom balanced asymmetry	2	1.689	99.8%	0.1%–5.4%	1.8%–7.2%	28.0%–
directional easy-out / hard-return	2	1.614	99.9%	0.1%–8.0%	1.3%–9.1%	27.8%–
directional hard-out / easy-return	2	1.737	99.8%	0.1%–2.8%	2.5%–5.2%	28.2%–

Table 5: Two-target representative cases.

case	phase	P_{near}	P_{far}	t_{p1}	t_v	t_{p2}	sep_{gate}
anchor	2	0.324	0.676	7	196	469	2.493
clear_instance	2	0.350	0.650	13	207	471	2.380
near_mass_loss	2	0.089	0.911	27	193	470	2.229

Table 6: Two-target validation.

case	$q_{far}(start)$	P_{far}	gap	closure	MC err
anchor	0.676009	0.676009	0.000000	0.000000	0.006816
clear_instance	0.650264	0.650264	0.000000	0.000000	0.004222
near_mass_loss	0.911262	0.911262	0.000000	0.000000	0.005743

Table 7: Phase counts on the 63-point $(d, \Delta y)$ atlas.

phase	count
0	2
2	61

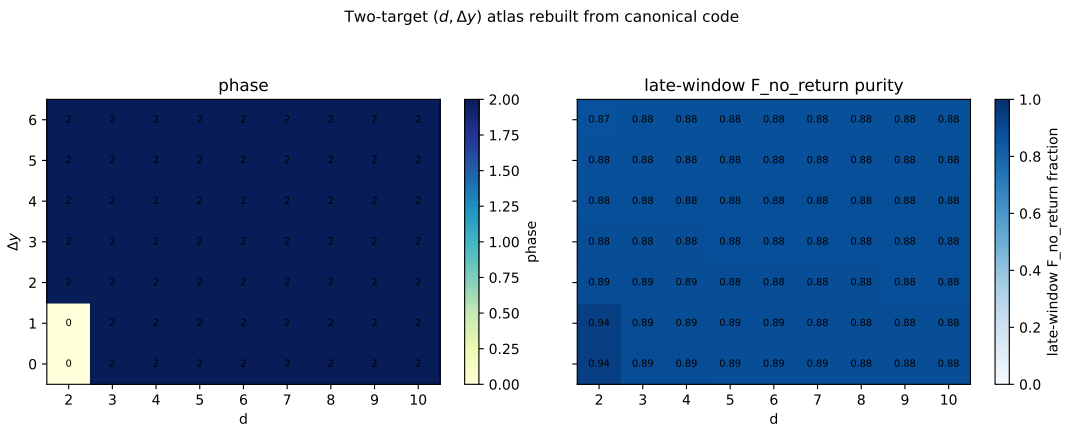


Figure 17: The 63-point two-target atlas. The current canonical rebuild still keeps phase-0 loss confined to a small subset of the grid.

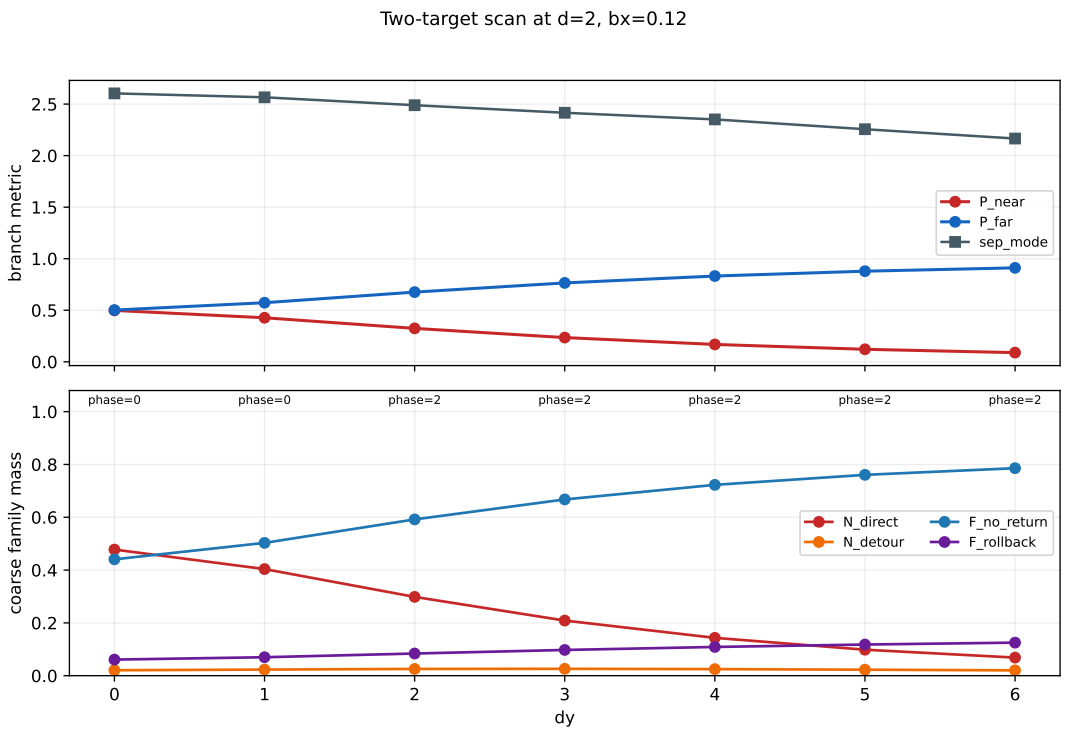


Figure 18: The Δy scan at fixed $d = 2$.

Table 8: Representative two-target recheck.

case	source P_{far}	recheck P_{far}	$q_{far}(start)$ gap	late family	MC err
anchor	0.676009	0.676009	0.000000	F_no_return	0.0068
clear_instance	0.650264	0.650264	0.000000	F_no_return	0.0042
near_mass_loss	0.911262	0.911262	0.000000	F_no_return	0.0057

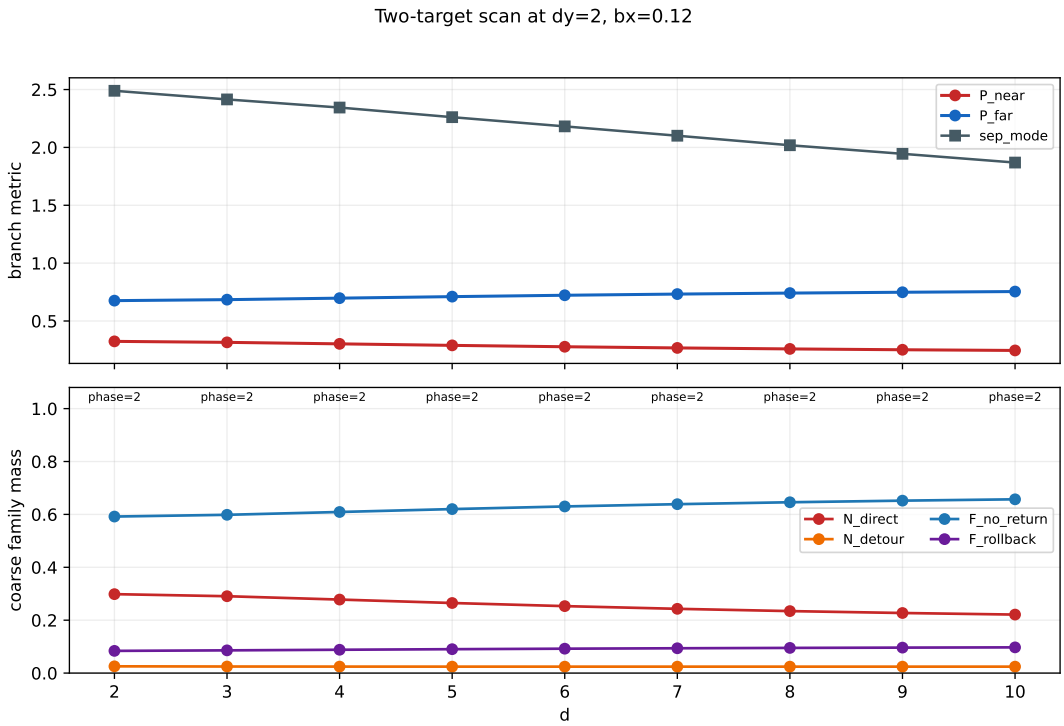


Figure 19: The d scan at fixed $\Delta y = 2$.

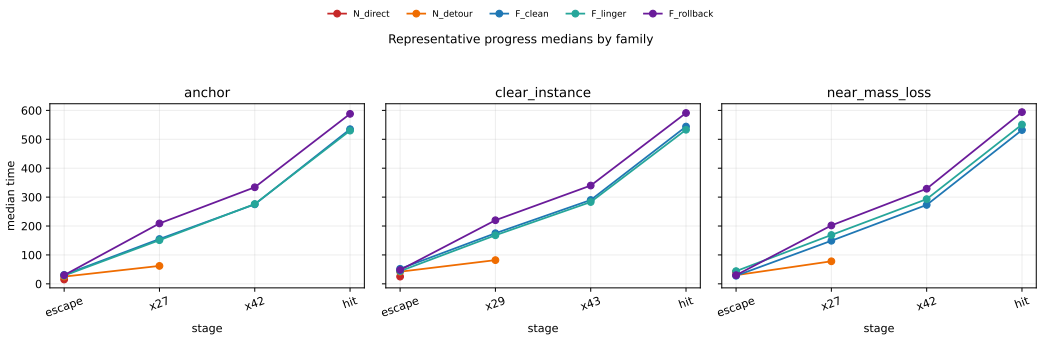


Figure 20: Representative progress medians.

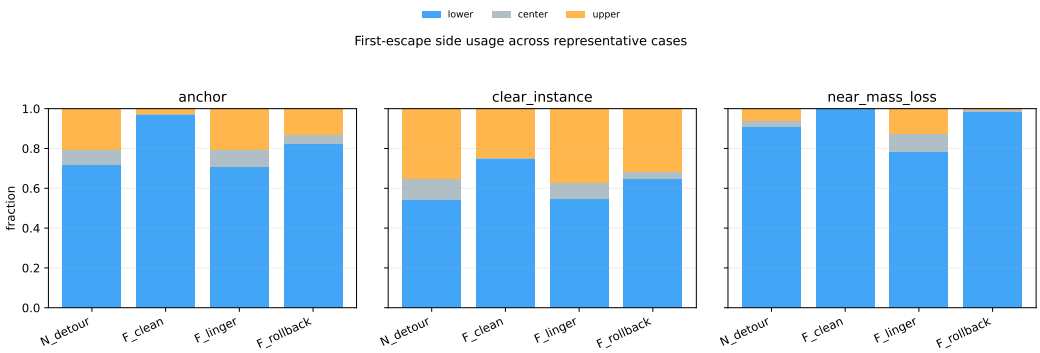


Figure 21: Representative first-escape side usage.

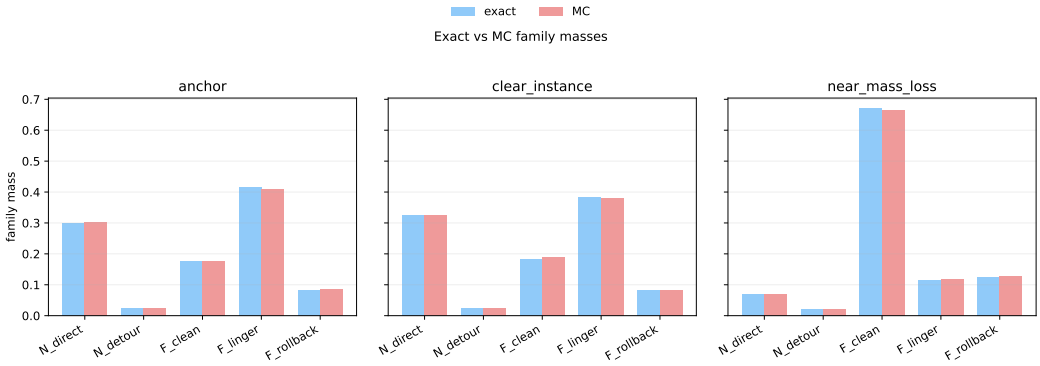


Figure 22: MC-vs-exact comparison.

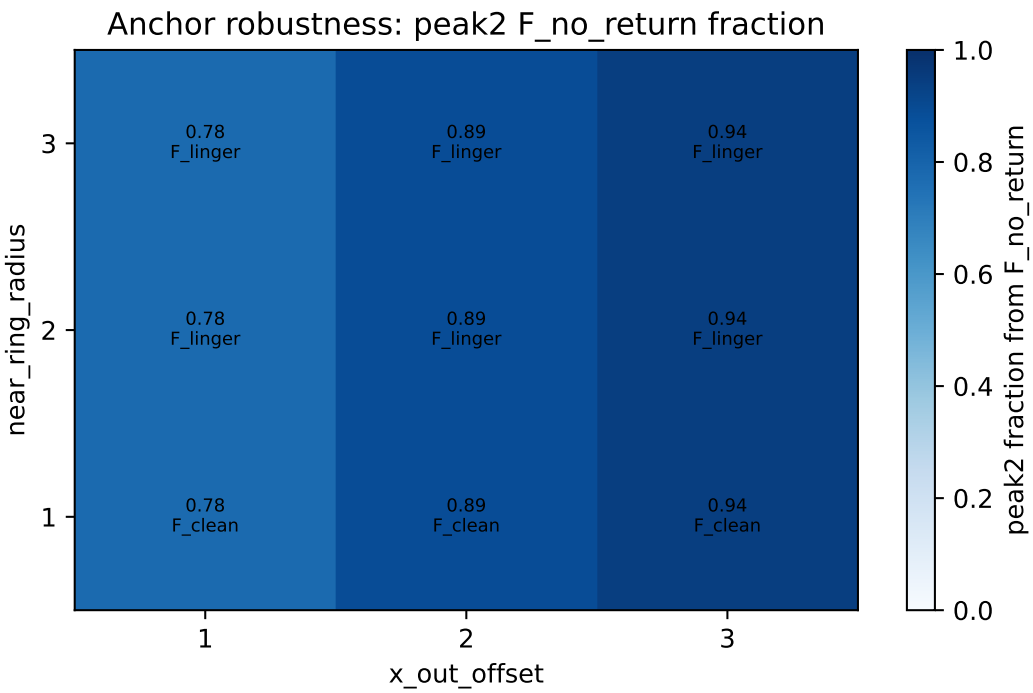


Figure 23: Gate robustness for the anchor case.

9 Repository integration and appendix note

Repository integration. The six original downloaded files remain archived under `notes/source_imports/202`
The new `artifacts/` and `manuscript/` trees are rebuilt directly from `vkcore.grid2d.one_two_target_gating.`

The final one-target language of this version. The one-target final language can now be summarized in one sentence:

> the standard configuration is the shared symmetric baseline; rollback is the main discrete mechanism; the x-gate is a timing anchor; bimodality loss means `phase=0`; and extreme $\kappa = 0$ points are controls rather than main representatives.

10 Appendix: committor control

The committor construction remains in the repository, but only as an appendix-level control. It is still useful for historical continuity and numerical sanity checks, yet it no longer defines the main one-target gate language.

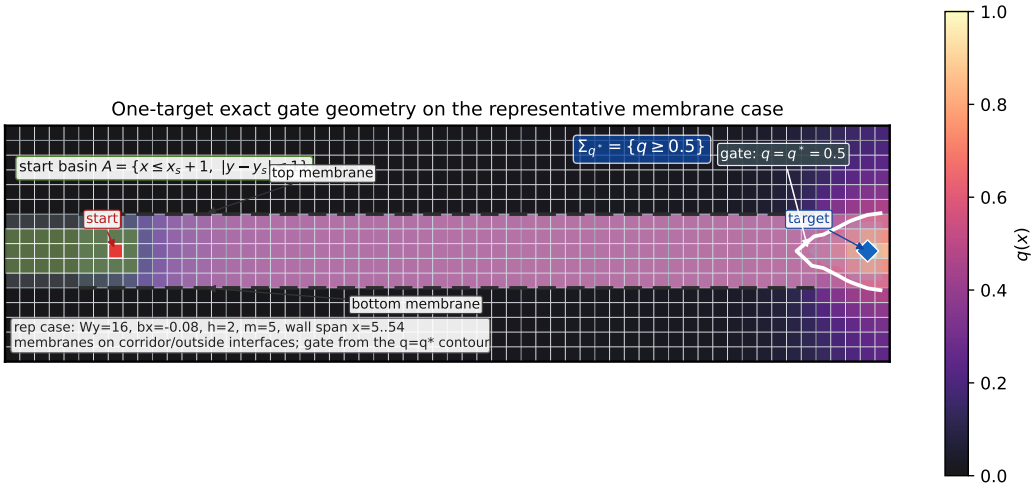


Figure 24: Committor-based control figure for the one-target model. The $q = q^*$ contour survives, but only as an appendix control.